

## Ghost Coast (Coastal / Ground + Naval + Air Operations) Map Design

### Executive Summary

Ghost Coast is a hybrid coastal battlefield designed for synchronized **amural, naval, and air operations**. The terrain features rugged cliffs, sandy beaches, offshore platforms, naval docks, submerged ruins, and elevated coastal roads. The map emphasizes Obidian Protocol's autonomous multi-domain tactics, sensor-fusion warfare, comm-net networks, and tactics across land, sea, and air. In AR (Real-Lite/VR), the map anchors at ~4-5 m viewing distance, covering a 20-30 m official area representative "1 km" of coastal territory. Key features include naval landing zones, offshore artillery positions, offshore industrial sites, underwater hazards, and multi-route amphibious assault paths.

### Design Goals & Gameplay

#### Multi-Domain Combat (Land + Sea + Air)

Ghost Coast supports multi-domain ground vehicles, infantry, naval craft, drones, and aircraft. Players coordinate amphibious landings, airstrikes, and coastal pushes.

#### Autonomy & Multi-Route Decision-Making

Autonomous AI chooses between beach landings, cliff ascents, underwater approaches, or aerial flank routes. Units adjust to tides, waves, and cliff-face visibility.

#### Sensor & Information Warfare

Sea mist, wave spray, and fog distort sensors. Radar stations on cliffs restore clarity. Destroying relay towers creates fog-of-war pockets across sea and land.

#### Landmarks & Resources

Naval docks, fuel depots, and offshore platforms serve as supply nodes. Long supply lines across open water and exposed beaches force careful convoy protection.

#### Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between naval landing craft, air support, or heavy ground armor. Blue displays from the southwest beachhead, Red displays from the northeast cliff fortress.

### Spatial Constraints (Nreal AR)

#### Device Capabilities

Target spec: "XR" (e.g., Quest 3), 1800x1800 per-eye resolution. Optimal viewing distance ~4-5 m.

#### Playable Area & Scaling

100m x 100m area, 1 km<sup>2</sup> coastal zone (1,000 units). Designed for tabletop AR.

#### Tracking & Occlusion

100m x 100m area, 1 km<sup>2</sup> coastal zone. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

### Layout Overview

#### Coordinate System

Key Locations

Blue HQ (Beachhead Command) — (0,0). 140x140 m Amphibious landing zone with naval craft and armored vehicles.

Red HQ (Cliff Fortress) — (100,100). 140x140 m fortified offshore base with radar mast and AA emplacements.

Central Coastal Town — (50, 50). 200x200 m District. Docks, shops, narrow streets, central ruins.

Naval Dockyard — (20, 80). 120x120 m Facilities. Cranes, cargo containers, supply boats.

Offshore Industrial Site — (80, 30). 150x150 m Platform. Ammunition, fuel tank, underwater supports.

Underwater Ruins — (40, 10) & (60, 90) Submerged structures, heavily overgrown rocks.

Cliff Road Network — (0,40) & (40,80) Elevated road with artillery positions.

Artillery Batteries — (10,10), (10,90), (40,10), (40,90) in Ruined towers, drone pads.

Radar Ridge — (10,50), (40,50) in Central ridge tower and sensor hub.

Neutral Forward Base — (50,50) 100x100 m base.

### Chokepoints & Sightlines

#### Beach Landing Zone

Open but exposed, naval air vulnerable during landing.

#### Cliff Road Network

Narrow elevated path, ideal for artillery ambushes.

#### Coastal Town Streets

Close-range combat, interior routes and alley flanks.

#### Offshore Rig

High ground for air units, naval units must approach carefully.

#### Sightlines

Radar Ridge provides long-range sensor coverage (~150 m). Sea mist and fog create fog-of-war pockets.

### Deployment Zones

Blue Deployment: 100 Beachhead — naval cluster area.

Red Deployment: 160 Cliff Fortress — AA emplacements.

Neutral Forward Base: 100x100 m 100x100 m base.

### Assets & Environment

#### Terrain

Cliffs, beaches, dunes, submerged ruins, coastal roads.

#### Vegetation

Sparse palms, shrubs, mangroves.

#### Buildings & Props

Military bunkers, AA guns, radar dishes, watchtowers. Industrial: Cranes, pipelines, generators, offshore rig structures. Logistics: Boats, trucks, crates, fuel drums, drones.

#### Effects

Sea mist, fog, wave spray, smoke plumes, wind gusts.

#### Audio

Waves, wind, distant machinery, radio chatter, signals.

### LOD & Budget

#### Asset Cate Count

Cliff/Cave 1 200k

Beaches/R 4 80k 20k

Buildings/ 15 15k 75k

Roads/Docks 4 10k 10k

Vegetation 100 2k 200k

Props 50 10k 500k

Total — — ~2.5M

#### Texture Budget

2048 for terrain/buildings, 1024 for props, 512x512 for compression. Target 1000 MB.

### Unreal / Nreal Integration

#### Engine & Plugins

Unreal ARComplete + Nreal SDK (NREAL). ARKit/ARCore enabled.

#### Rendering

Forward rendering, stationary lights, baked GI, aggressive culling.

#### Anchor

Single spatial anchor for entire map.

#### Interaction

Phone gesture for pan/zoom, HUD as screen-space widgets.

#### Multitasker

Client-server architecture, offload AI/placement to server.

### Implementation Plan & Testing

#### Milestones

1. Prototype Level — Month 1 Block-out cliffs, beaches, offshore rig, AR anchor test.

2. Core Systems — Month 2 Beachhead, naval/artillery autonomous behaviors.

3. AR Interaction — Month 3 Anchors, scene detection, basic interactivity.

4. Content Fill — Month 4-5 Final models, textures, lighting, audio.

5. Gameplay Polish — Month 6-7 Obidian, balance, VR/AR mod, fix, wave coral.

6. Testing & Opt — Month 8 Multitasker stress test, sensor validation.

### Testing Plan

#### Autonomy

VR/AR multi-lane test, aerial flanking, underwater pathfinding.

#### Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

#### Performance

400 Hz AR render on target device.

#### QA Checklist

Navigation, multi-domain coordination, deployment budget enforcement, anchor drift, sensor accuracy.